



US012416310B2

(12) **United States Patent**  
**Chang et al.**

(10) **Patent No.:** **US 12,416,310 B2**  
(45) **Date of Patent:** **Sep. 16, 2025**

(54) **MAGNETICALLY ATTRACTED FAN SET**

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(\*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **18/357,822**

(22) Filed: **Jul. 24, 2023**

(65) **Prior Publication Data**

US 2023/0366407 A1 Nov. 16, 2023

**Related U.S. Application Data**

(62) Division of application No. 17/329,656, filed on May 25, 2021.

(51) **Int. Cl.**  
**F04D 25/16** (2006.01)  
**F04D 25/08** (2006.01)  
**F04D 29/40** (2006.01)  
**F04D 29/64** (2006.01)

(52) **U.S. Cl.**  
CPC ..... **F04D 25/166** (2013.01); **F04D 25/08** (2013.01); **F04D 29/403** (2013.01); **F04D 29/646** (2013.01); **F05D 2260/32** (2013.01)

(58) **Field of Classification Search**  
CPC ..... **F05D 2260/32**; **F05B 2260/302**  
See application file for complete search history.

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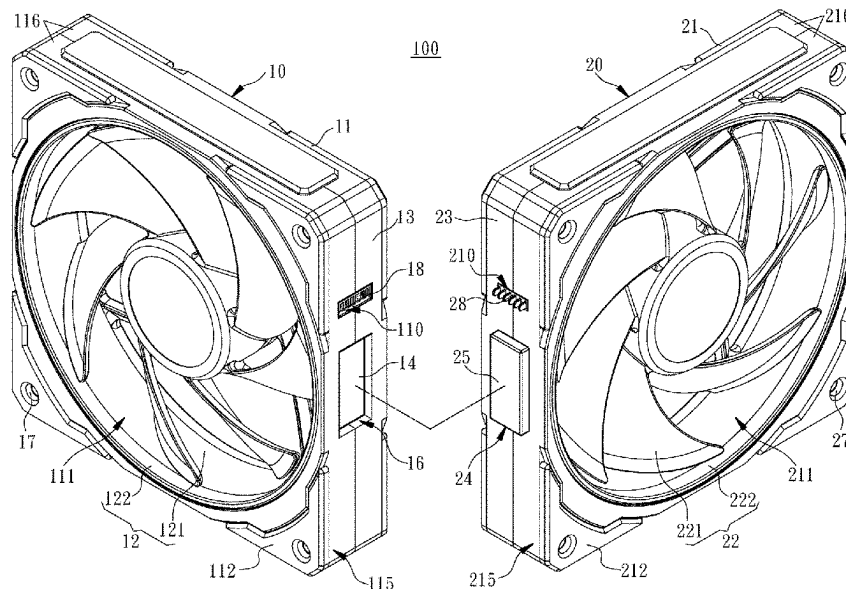
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(57) **ABSTRACT**

A magnetically attracted fan set comprises N+1 fans, the  $N \geq 1$ , and each of the N+1 fans has a fan frame and an airflow generating component disposed on the fan frame. Any two adjacent fans of the N+1 fans respectively have a joint surface, each of the joint surfaces has at least one fixing member capable of forming magnetic attraction, when the joint surface is close to the other joint surface at a distance sufficient for the two fixing members on the two joint surfaces to form magnetic attraction, two of the N+1 fans are joined together. Accordingly, the invention is capable of directly and quickly joining the N+1 fans without requiring complicated assembly gestures or other external components.

**17 Claims, 17 Drawing Sheets**



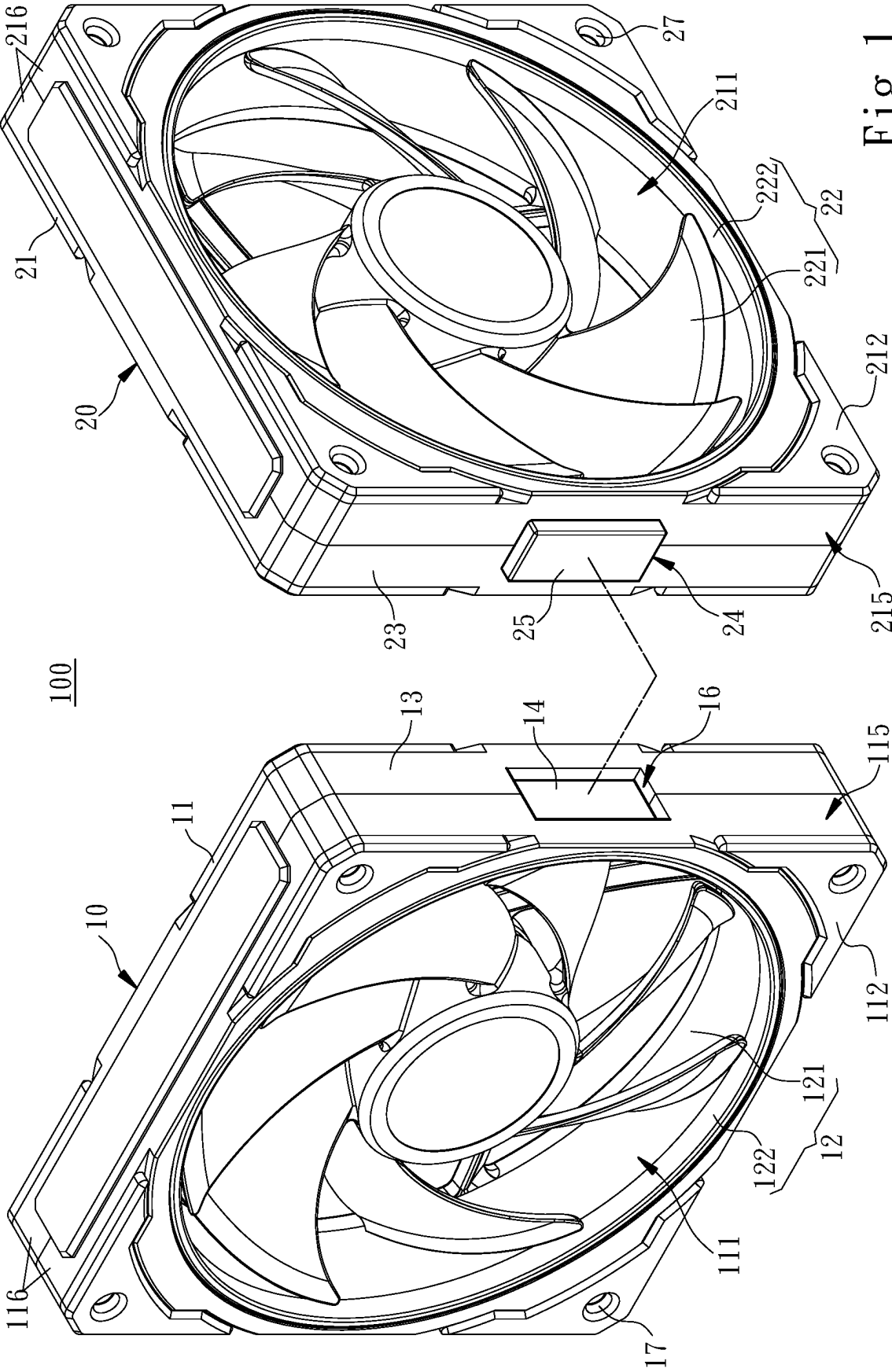


Fig. 1

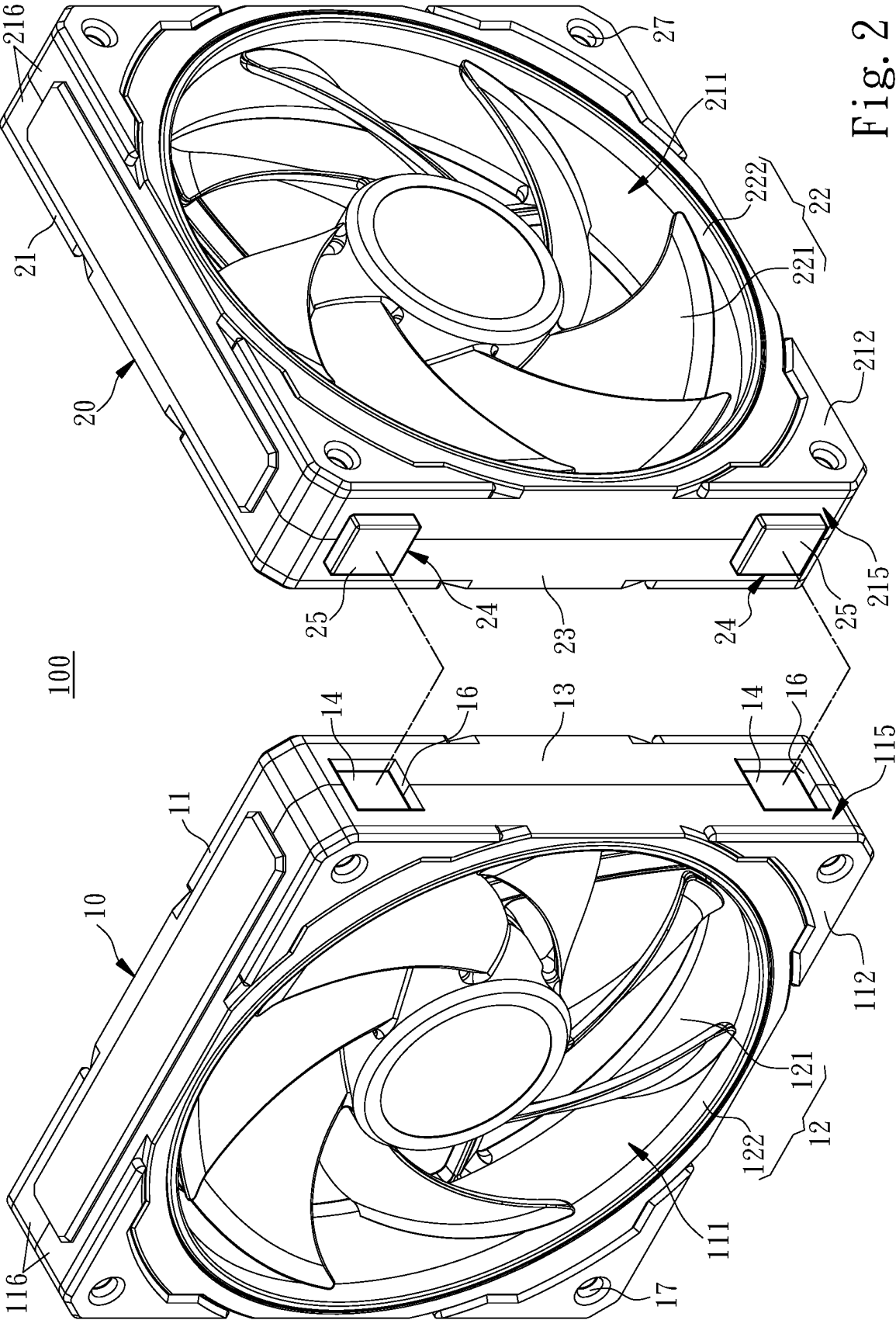


Fig. 2

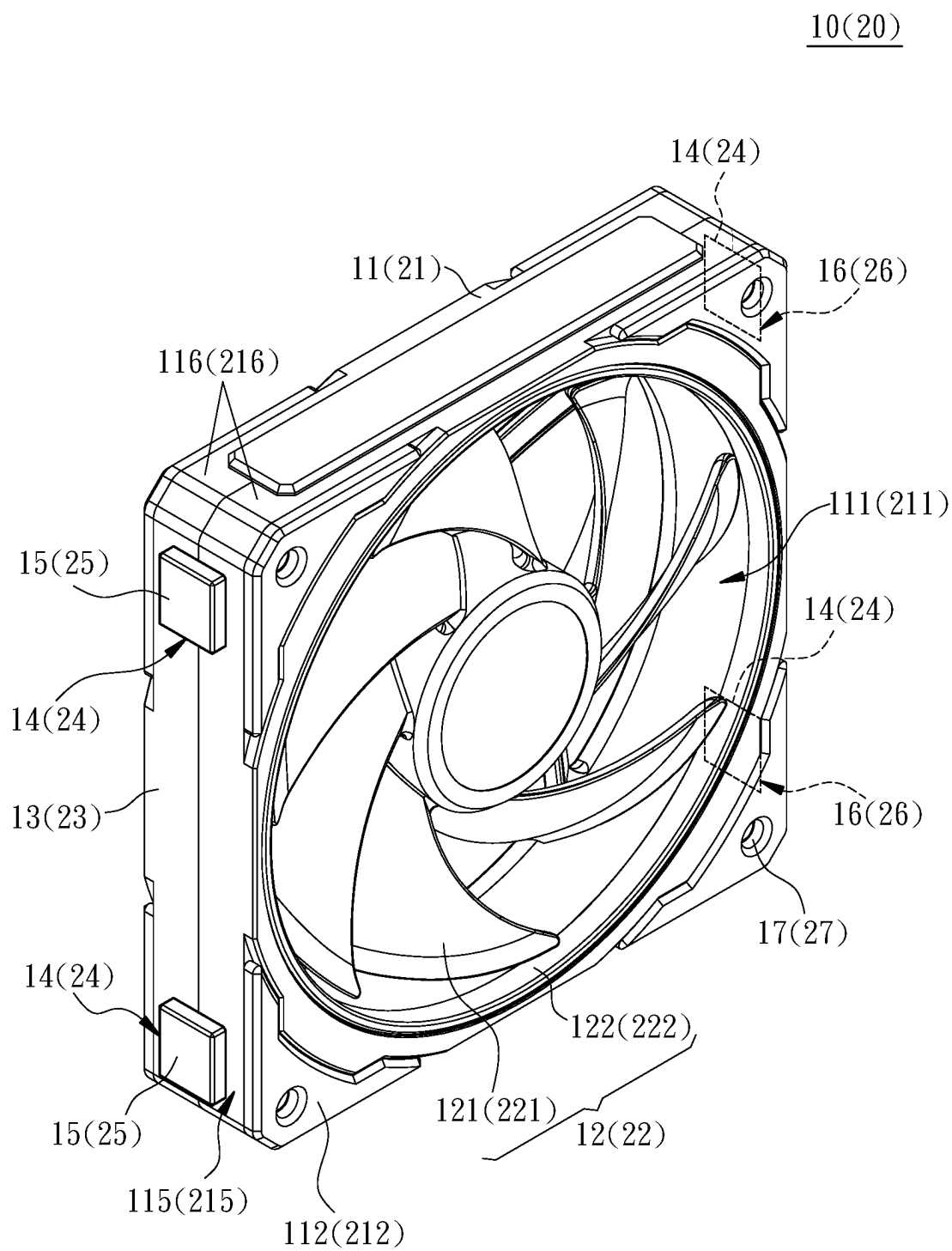


Fig. 3

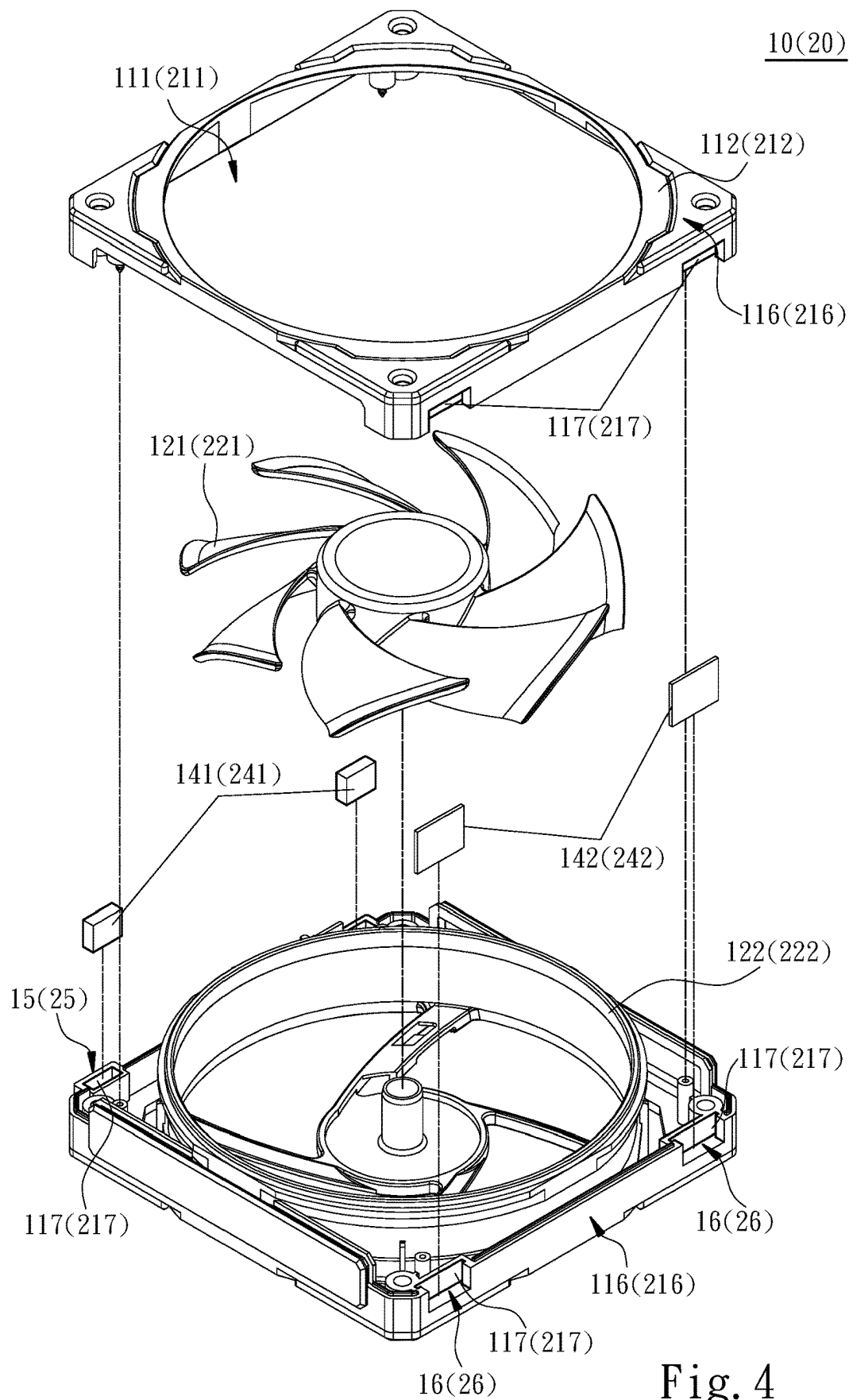


Fig. 4

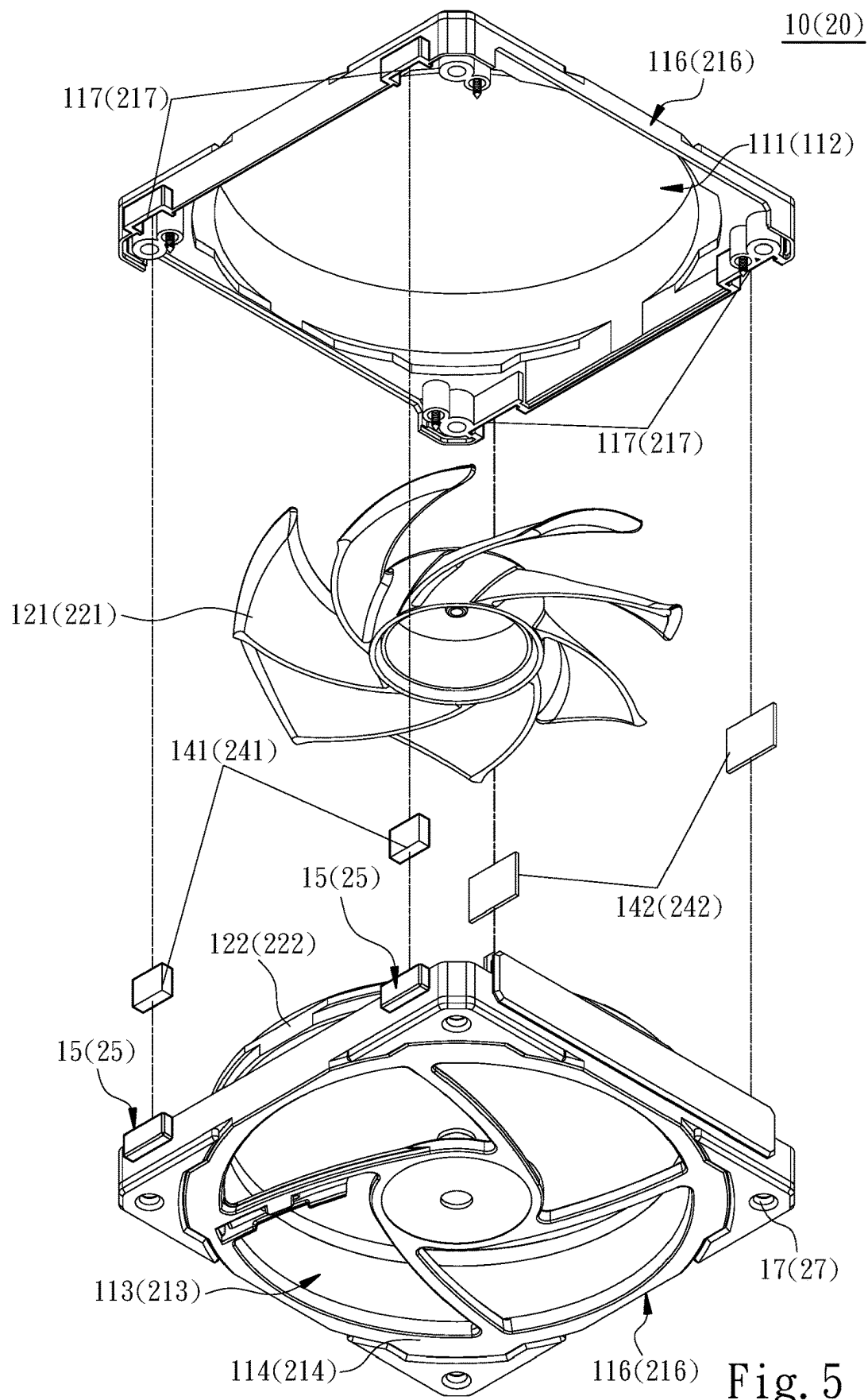


Fig. 5

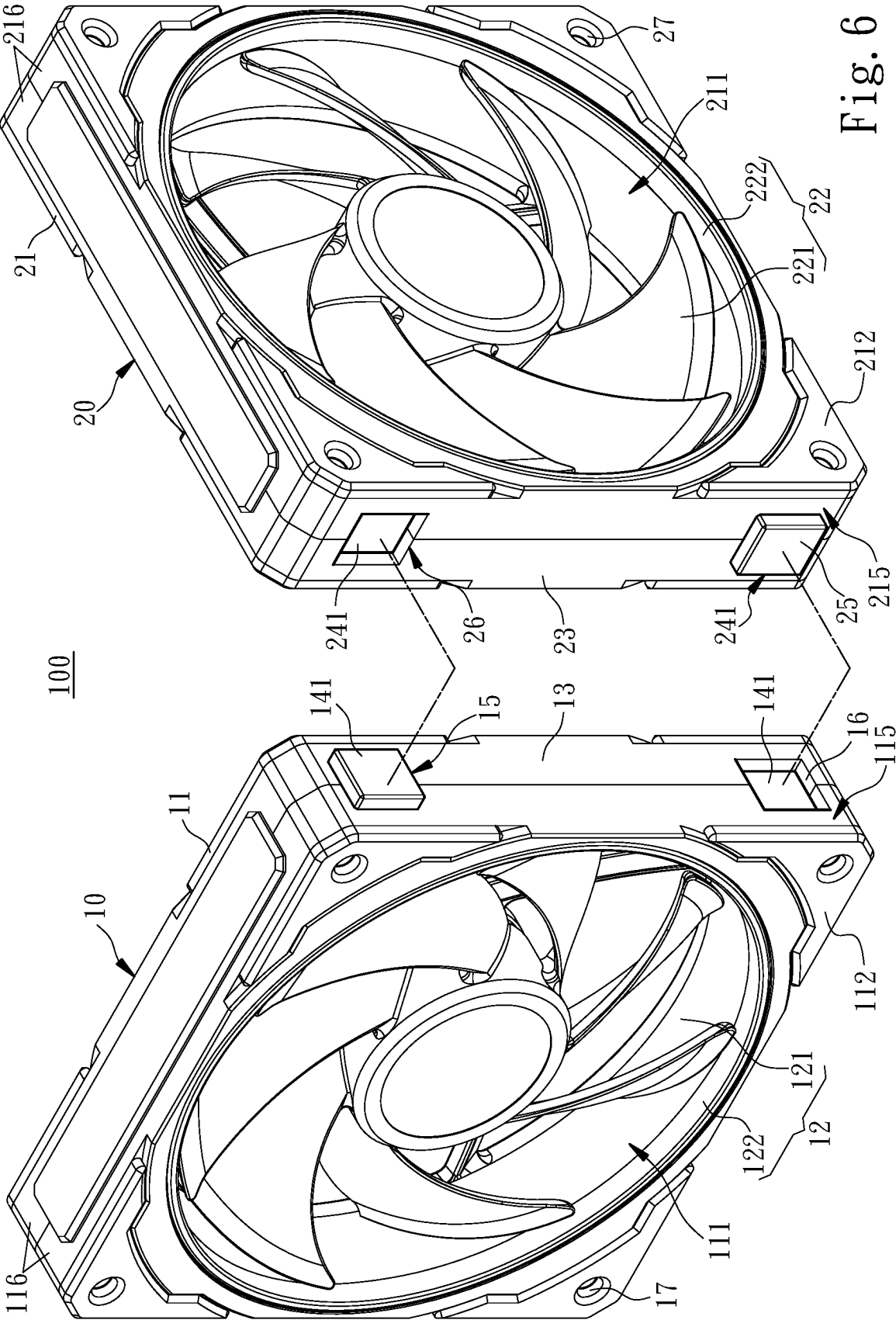


Fig. 6

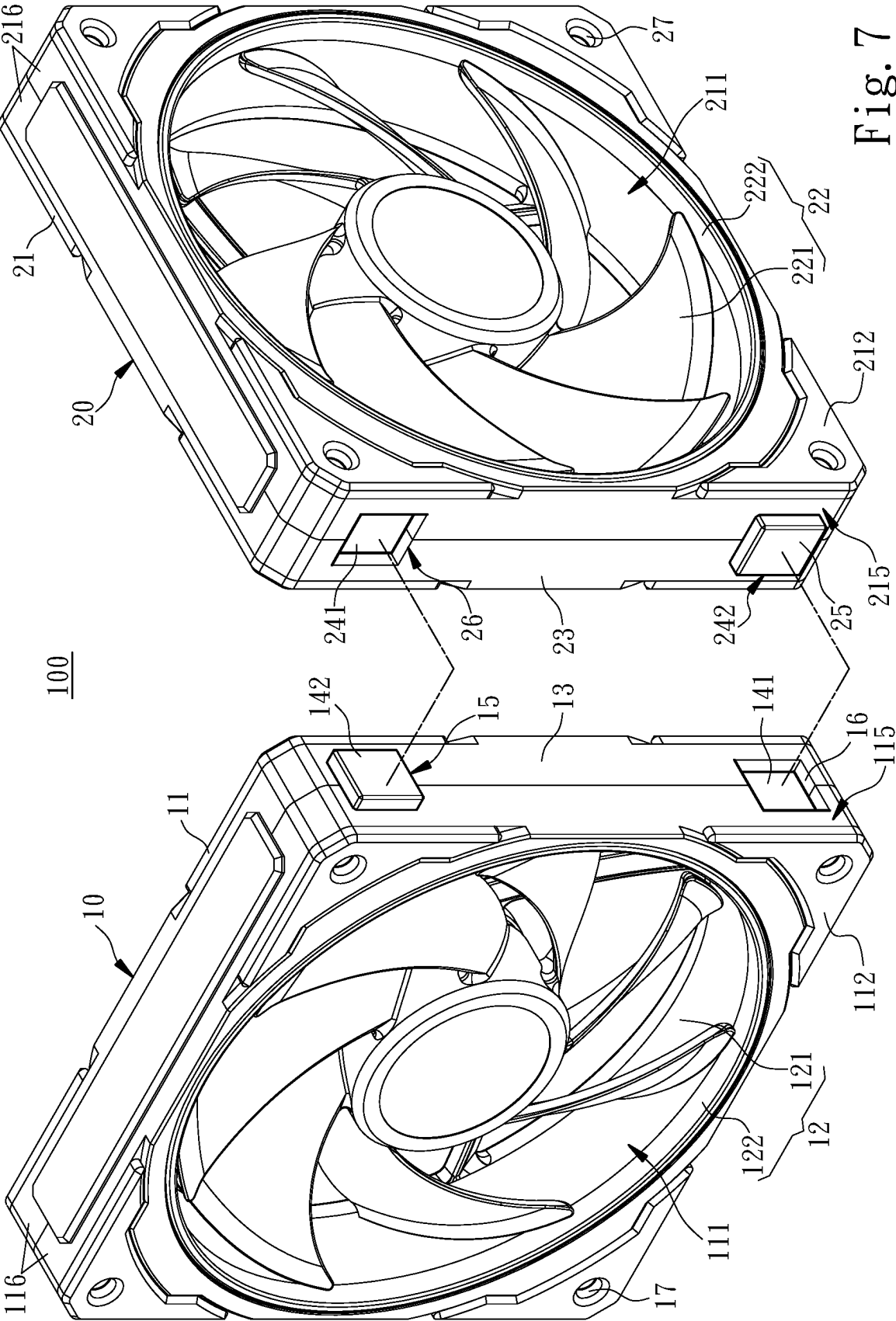


Fig. 7



100

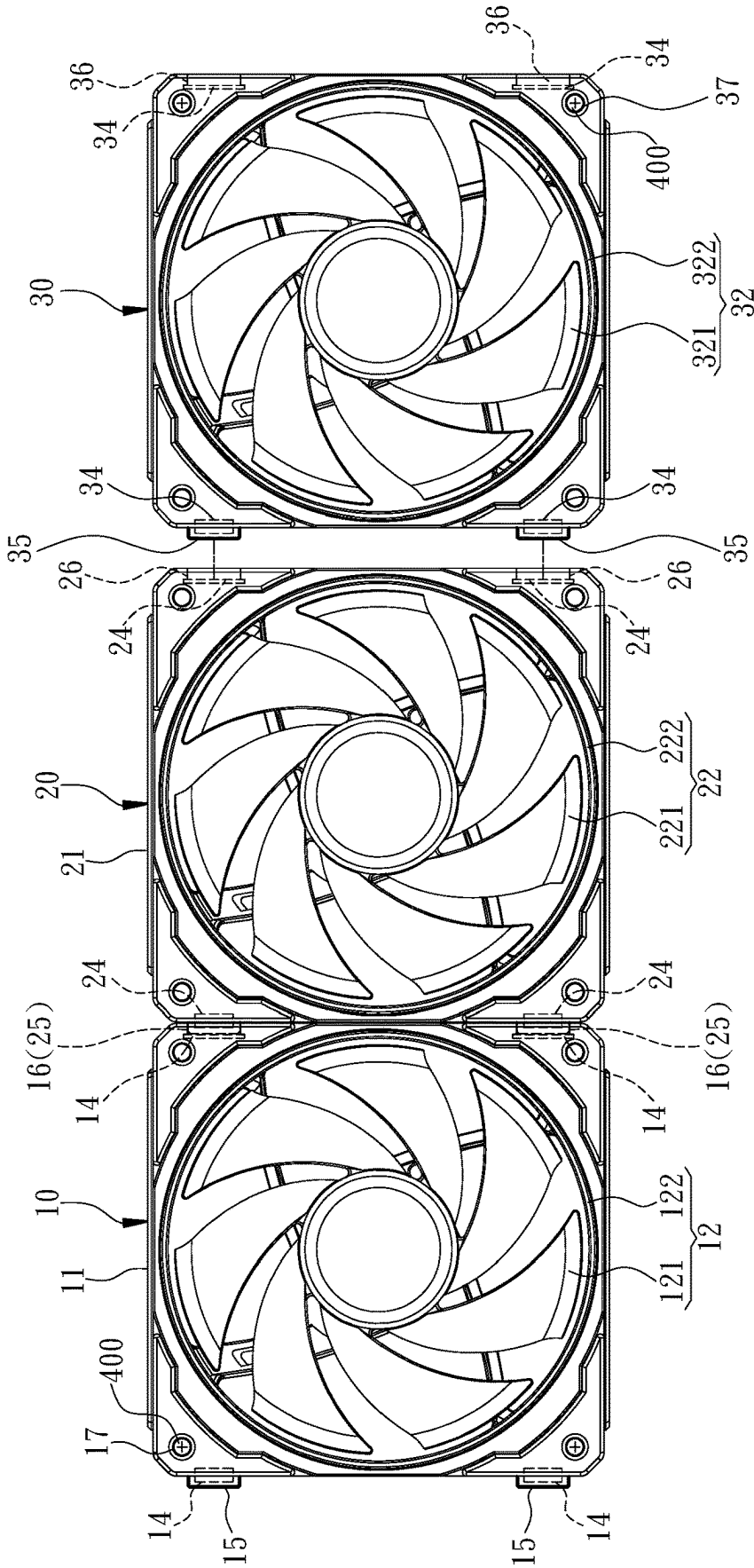


Fig. 8

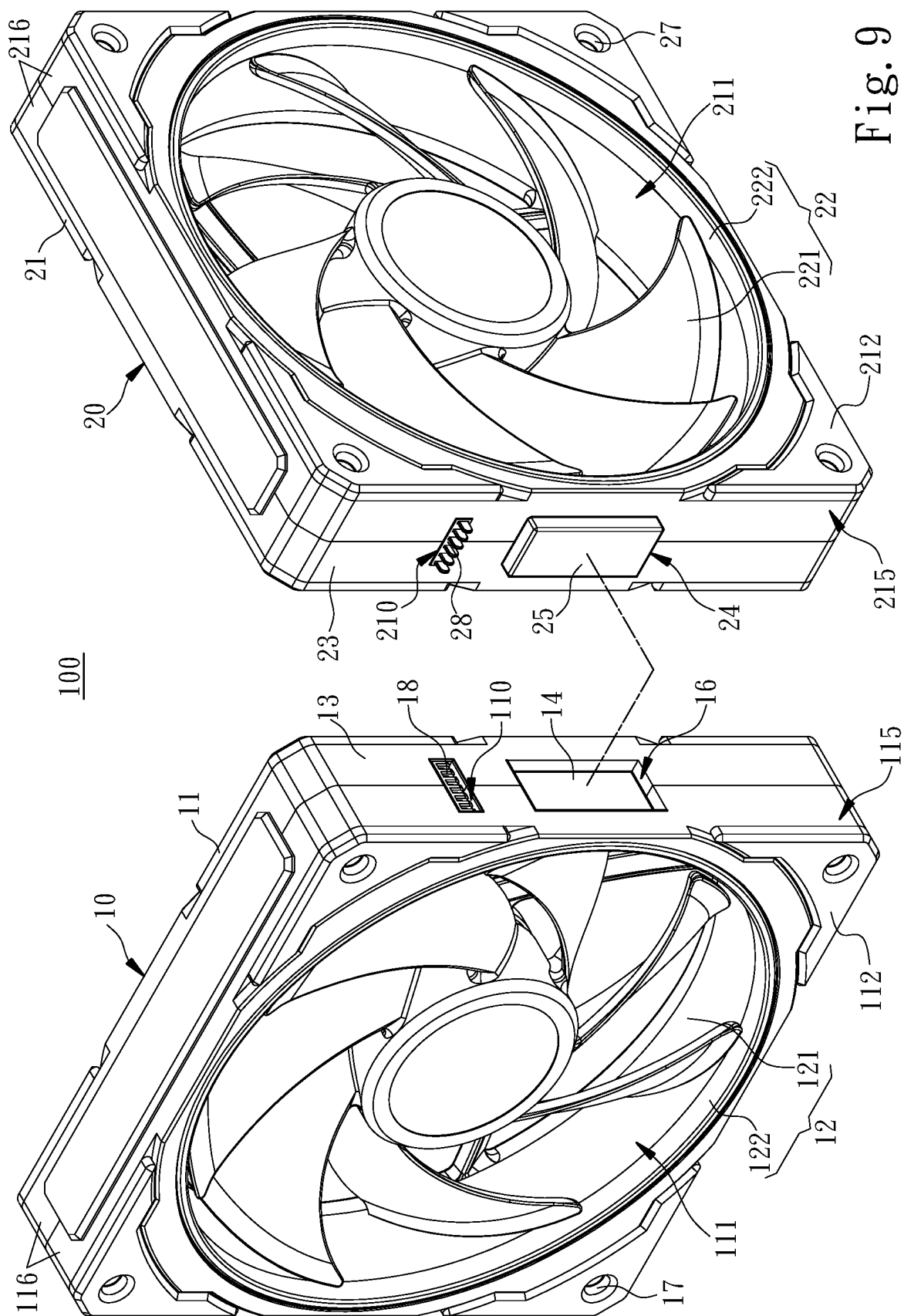


Fig. 9

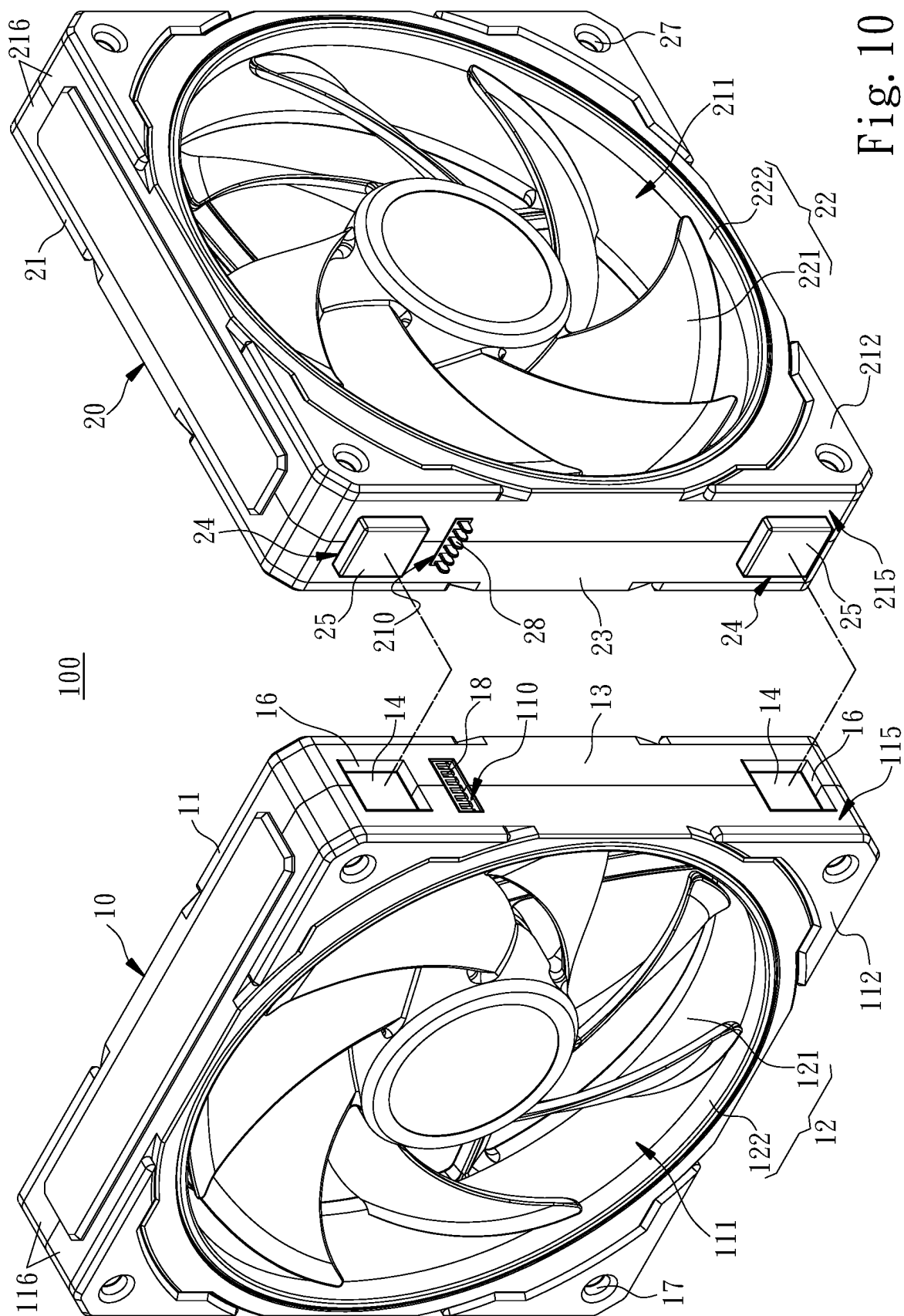


Fig. 10

10(20)

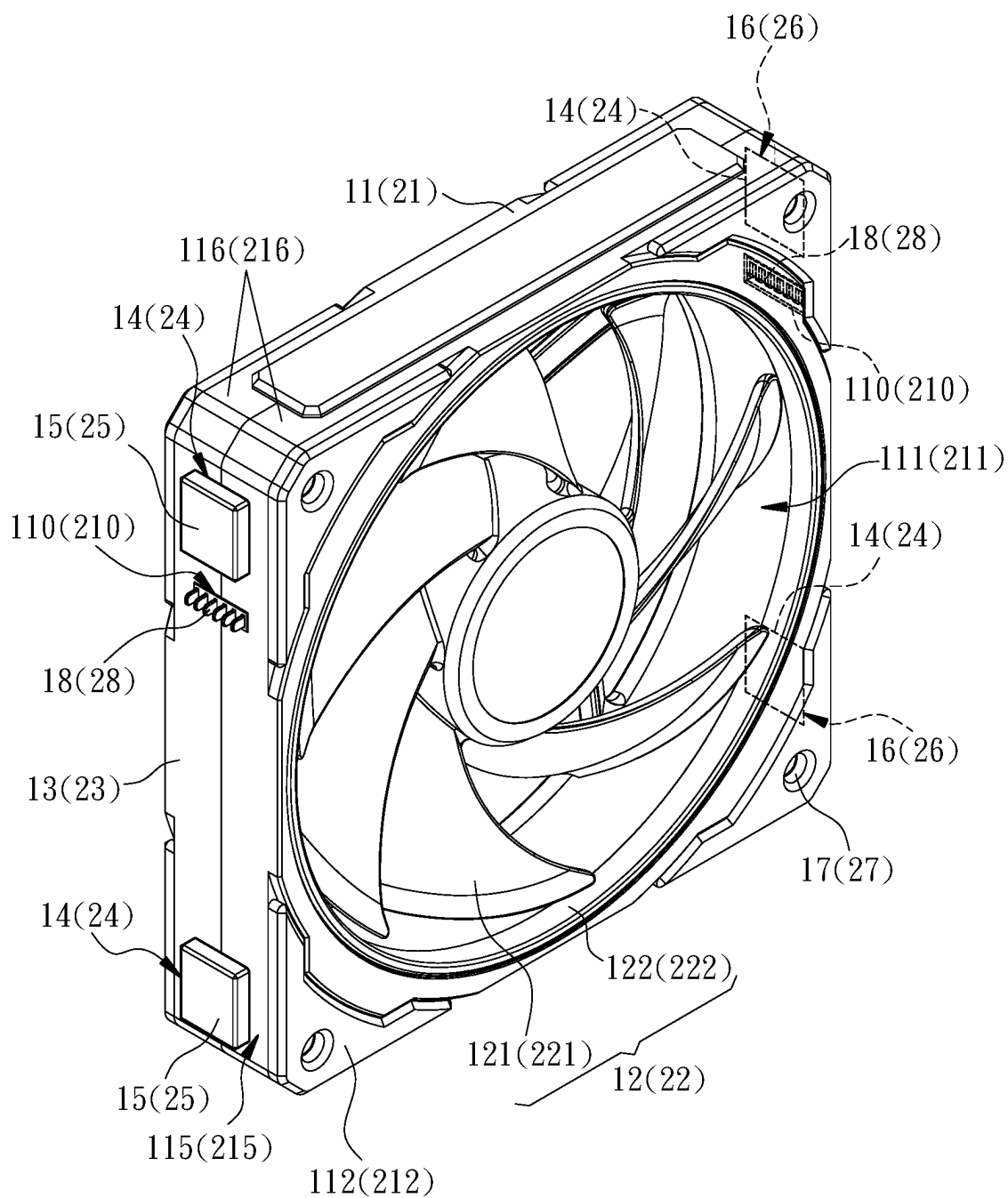
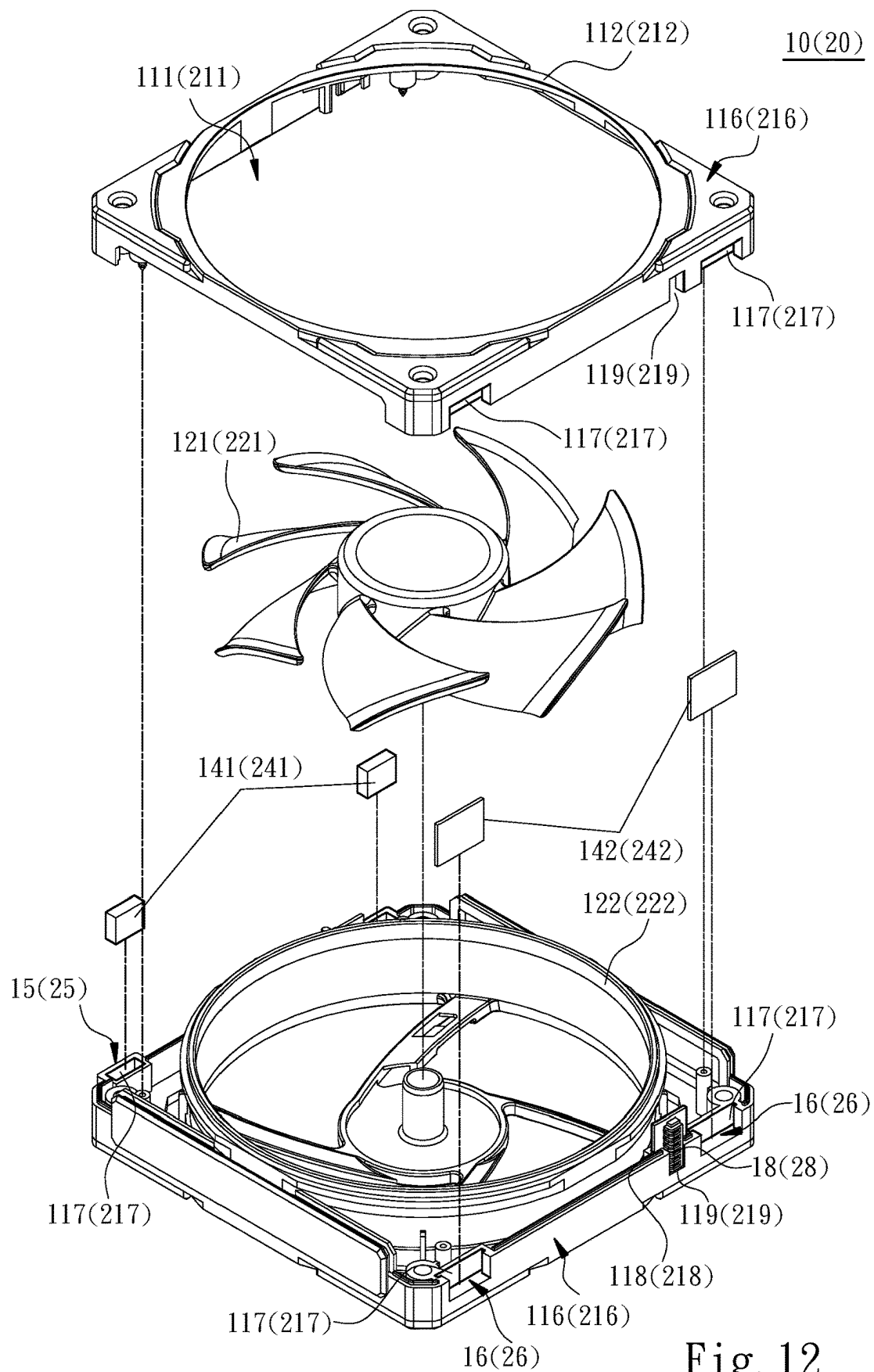


Fig. 11



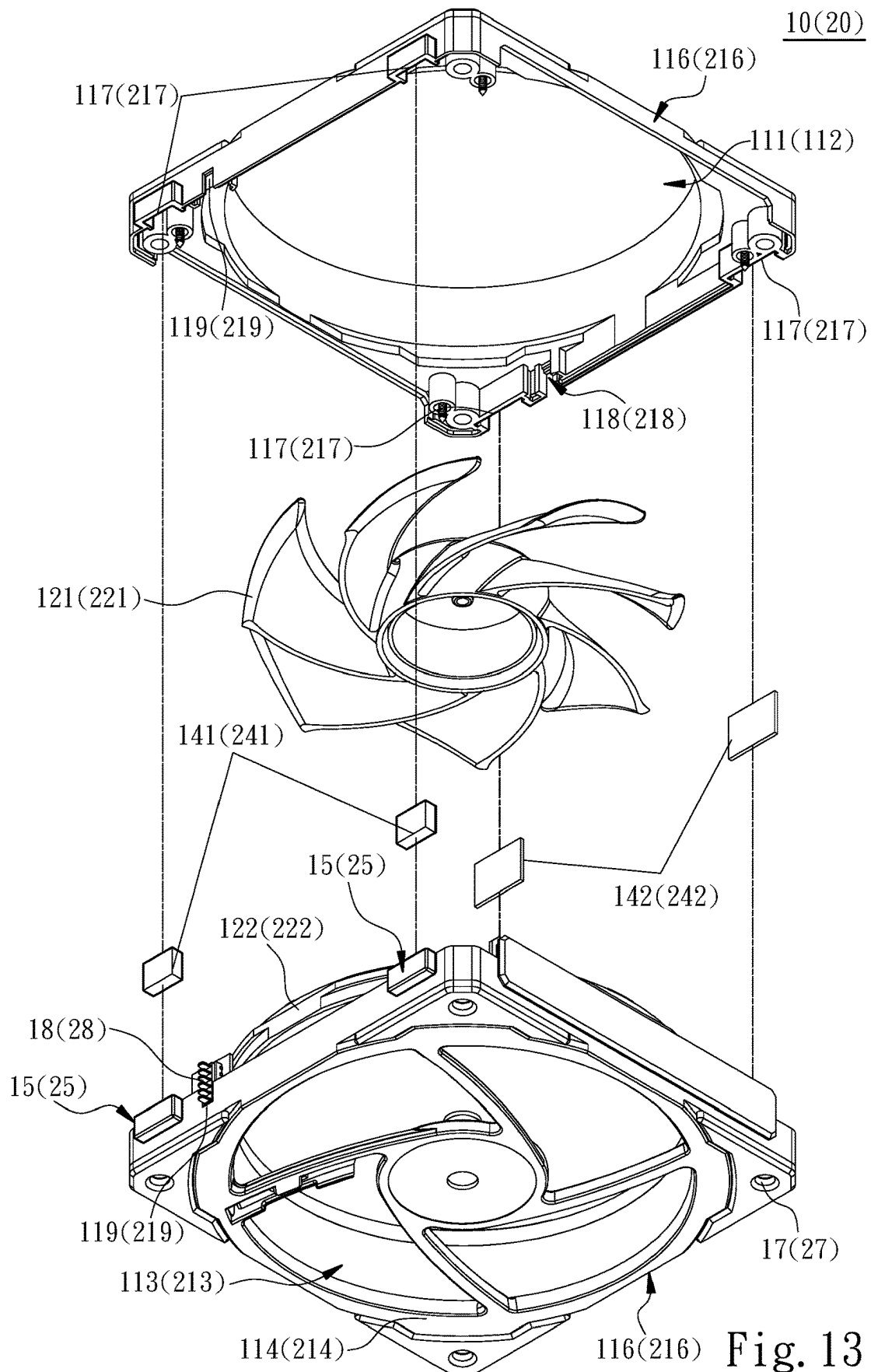


Fig. 13

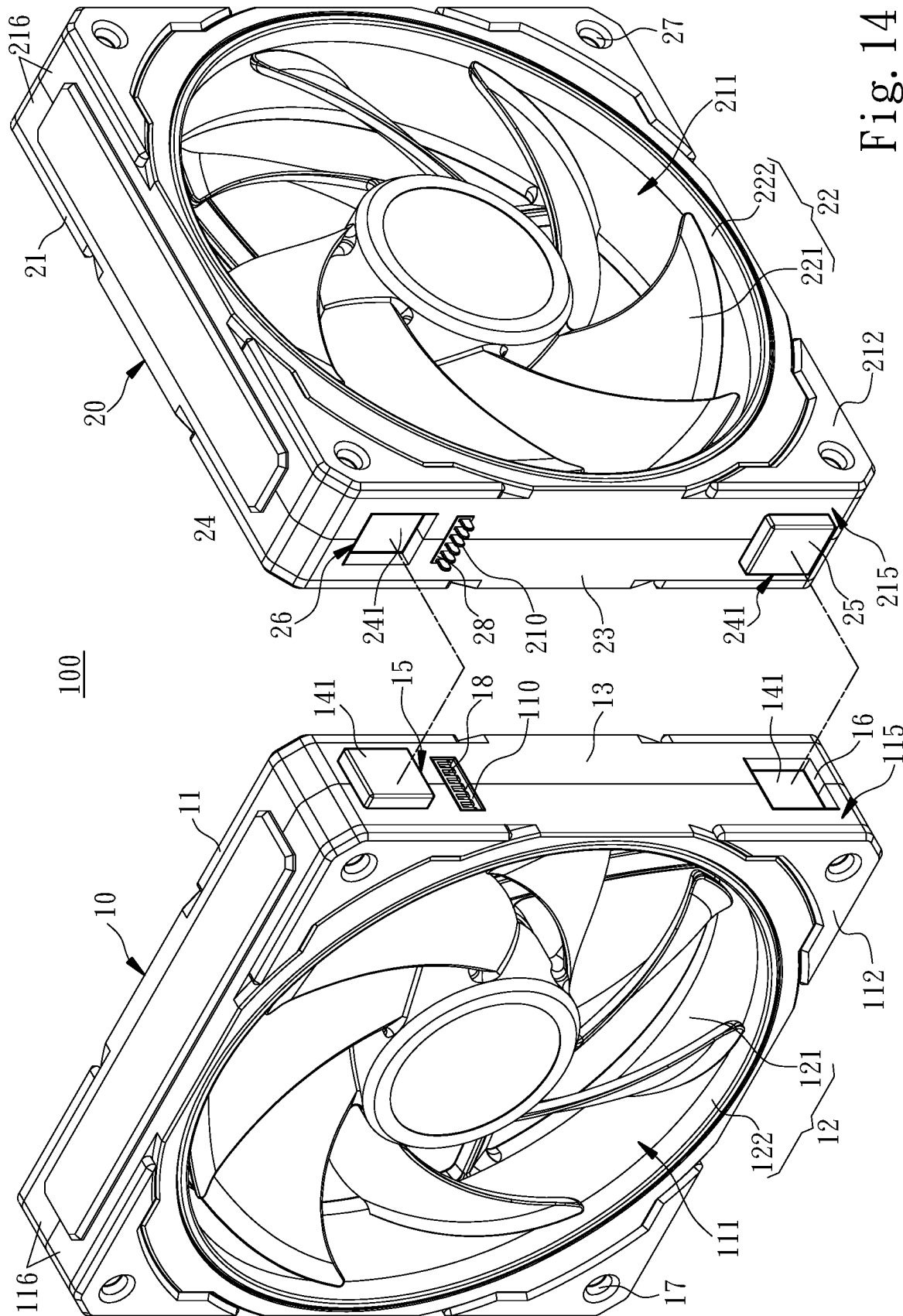


Fig. 14

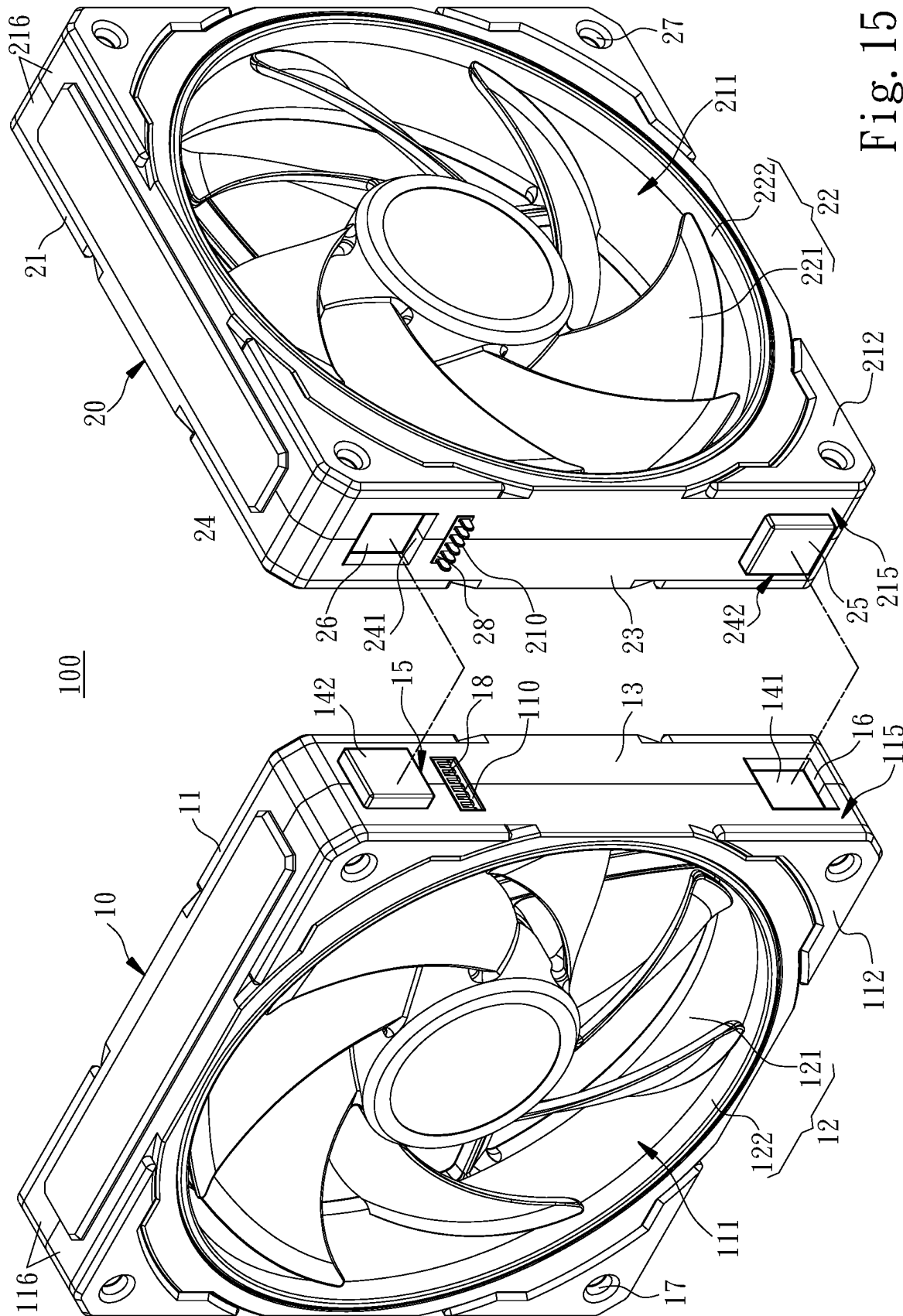


Fig. 15



100

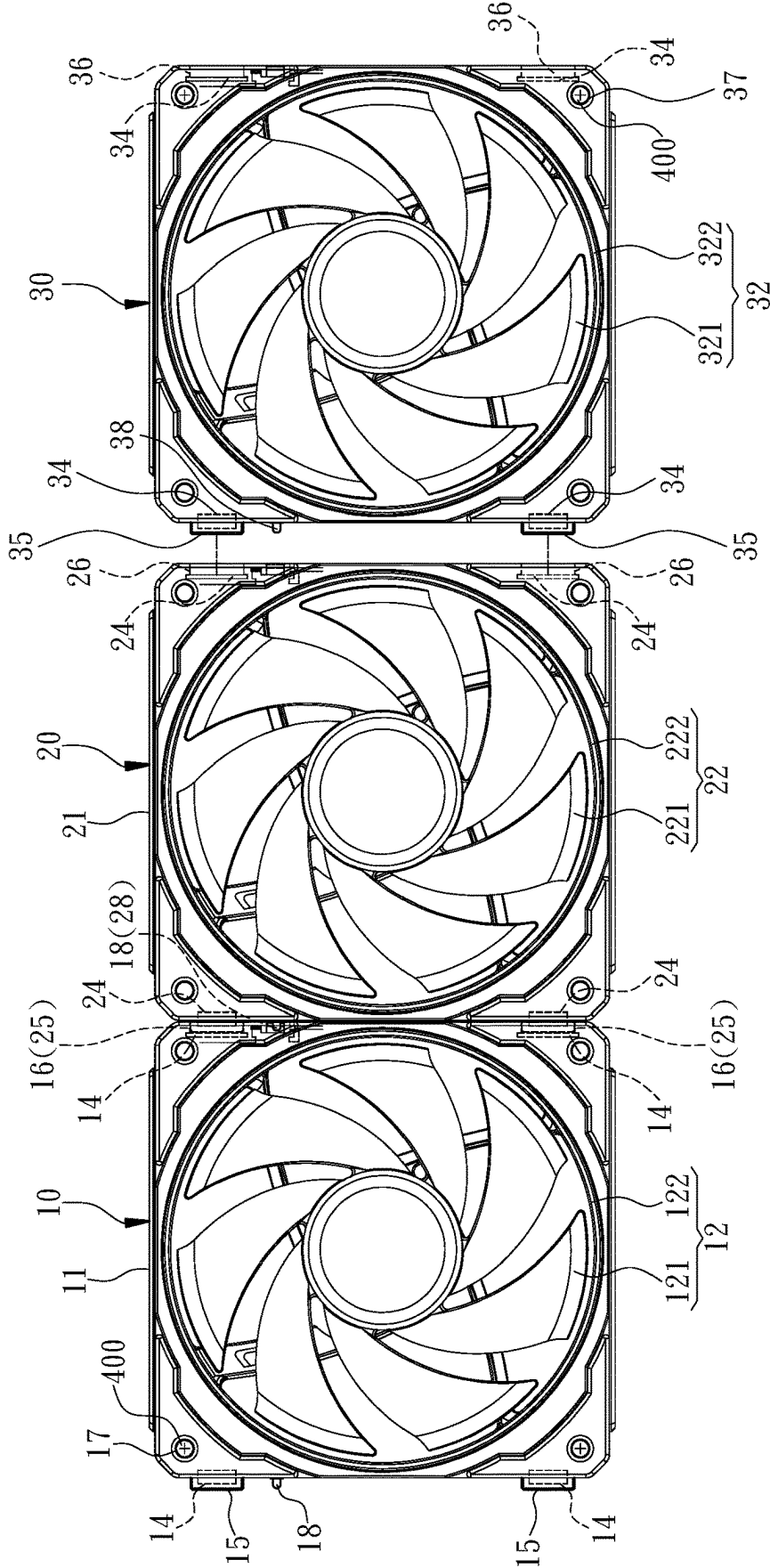


Fig. 16

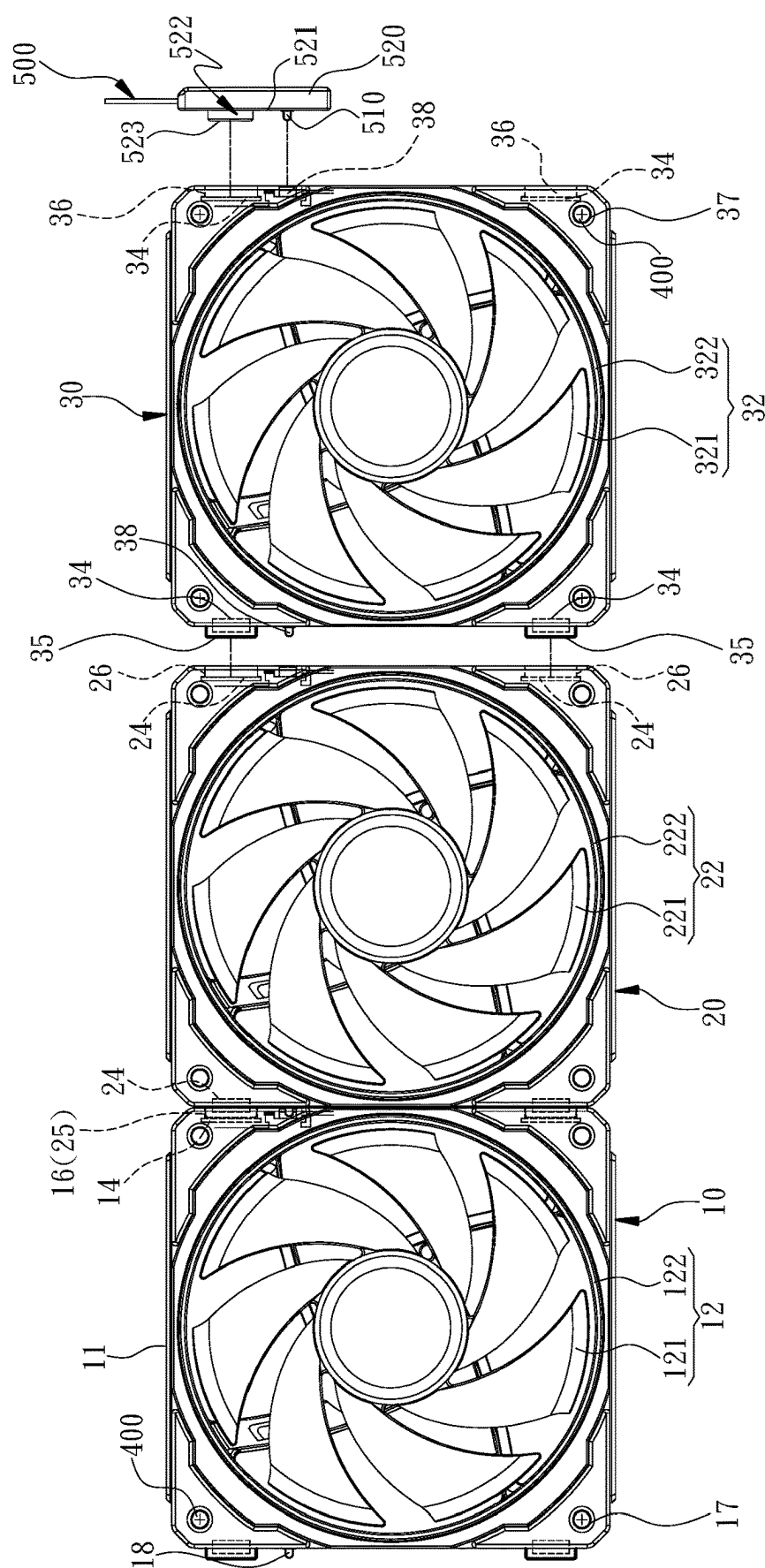


Fig. 17

**MAGNETICALLY ATTRACTED FAN SET****CROSS REFERENCE TO RELATED APPLICATIONS**

This application is a Divisional of co-pending application Ser. No. 17/329,656 filed on May 25, 2021, for which priority is claimed under 35 U.S.C. § 120; and the entire contents of all of which are hereby incorporated by reference.

**FIELD OF THE INVENTION**

The invention relates to a fan set, and more particularly to a fan set capable of joining together through magnetic attraction.

**BACKGROUND OF THE INVENTION**

According to investigation, currently, fan manufacturers have designed the fan frame in a way that can meet the user's demand for joining multiple fans in a horizontal manner. For example, TW M396884 and US 2017/0331346 disclose the technical means of joining fans through additional fixing members. Specifically, the fan frame is provided with a coupling groove or a coupling member, and the coupling groove and the coupling member are respectively provided with perforations in which the fixing member is disposed. When joining the fans, the user inserts the coupling member on one of the fans into the coupling groove on the other fan, so that the perforation on the coupling member and the perforation on the coupling groove are aligned, and then the fixing member is penetrated into the perforations, so that the fixing member restricts the coupling member and the coupling groove. However, the aforementioned means for joining multiple fans rely heavily on the limiting function provided by the fixing member, if the user is unable to use the fixing member when joining the fans, the fans cannot be fixed, and displacement is likely to occur. In addition, the conventional fixing member is an independent component, and in general, the size of the fixing member is significantly smaller than the fans, and therefore the fixing member can be lost easily during storage, causing inconvenience to the user.

In order to improve the aforementioned problems, fan manufacturers have changed to join the fans without using additional fixing members, as disclosed in patents such as U.S. Pat. No. 10,690,336B and CN 101725571B. Taking U.S. Pat. No. 10,690,336B as an example, the fans are provided with slots or positioning blocks on the fan frame. When joining the fans, the user connects and assembles one of the fan's positioning block to the other fan's slot to form an assembling relationship between the fans. However, the conventional technical solution of directly forming mechanical connection structures on the fans is likely to cause the mechanical connection structures to be misaligned due to excessive dimensional tolerances between the fans. Furthermore, the conventional fan has certain gestures in assembling or disassembling, which makes assembly and disassembly of the conventional fans complicated and inconvenient for users.

**SUMMARY OF THE INVENTION**

A main object of the invention is to solve the problem of inconvenience in assembling conventional fans.

In order to achieve the above object, the invention provides a magnetically attracted fan set comprising  $N+1$  fans,  $N \geq 1$ , and each of the  $N+1$  fans includes a plurality of fixing members for joining, and a plurality of transmission electrodes respectively located on a surface as same as at least one of the plurality of fixing members, the  $N+1$  fans are joined with one another through magnetic attraction generated by the plurality of fixing members, one of the plurality of transmission electrodes of each of the  $N+1$  fans is connected to one of the plurality of transmission electrodes of the other one of the  $N+1$  fans to transmit electric power. The control source includes a connecting seat, a control source electrode disposed on the connecting seat, and a fixing member disposed on a side of the connecting seat provided with the control source electrode, and generates a magnetic attraction effect is generated by the fixing member, when the fixing member and the plurality of fixing members of one of the  $N+1$  fans generate a magnetic attraction effect, the control source electrode is connected to one of the plurality of transmission electrodes.

In one embodiment, each of the plurality of transmission electrodes is disposed of close to one of the plurality of fixing members.

In one embodiment, each of the  $N+1$  fans includes two transmission electrodes, and the two transmission electrodes are respectively located on two sides of each of the  $N+1$  fans opposite to each other.

In one embodiment, the plurality of fixing members on one surface are located at two positions of the surface opposite to each other.

In one embodiment, the  $N+1$  fans are connected in series side by side.

In one embodiment, the  $N+1$  fans include a fan frame provided for disposing of the plurality of fixing members and the plurality of transmission electrodes, the fan frame includes a top plate including a first air opening, a bottom plate parallel to the top plate and including a second air opening communicating with the first air opening, and a side wall connecting the top plate and the bottom plate, the side wall is formed with a plurality of grooves provided for disposing of the plurality of fixing members and the plurality of transmission electrodes.

In one embodiment, a surface of the connecting seat provided with the fixing member contacts the fan frame flatly.

In addition to the above, the invention further provides another magnetically attracted fan set comprising  $N+1$  fans and a control source.  $N \geq 1$ . The  $N+1$  fans include a fan frame forming a plurality of grooves on two surfaces opposite to each other, wherein each of the two surfaces is without an air opening formed, the  $N+1$  fans include a plurality of fixing members disposed in the plurality of grooves, and at least one of the plurality of transmission electrodes disposed in one of the plurality of grooves, the  $N+1$  fans are joined with one another through a magnetic attraction effect generated by the plurality of fixing members, and the  $N+1$  fans transmit electric power through the plurality of transmission electrodes. The control source includes a connecting seat, a control source electrode disposed on the connecting seat, and a fixing member disposed on the connecting seat, the fixing member of the control source generates a magnetic attraction effect with one of the plurality of fixing members on one of the  $N+1$  fans, when the fixing member of the control source attracts one of the plurality of fixing members on one of the  $N+1$  fans, the control source electrode is connected with one of the plurality of transmission electrodes.

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In one embodiment, each of the plurality of transmission electrodes is disposed close to one of the plurality of fixing members.

In one embodiment, the plurality of fixing members on one surface are located at two positions of the surface opposite to each other.

In one embodiment, the N+1 fans are connected in series side by side.

In one embodiment, a surface of the connecting seat provided with the fixing member contacts the fan frame flatly.

In addition to the above, the invention further provides another magnetically attracted fan set comprising N+1 fans and a control source. The  $N \geq 1$ , the N+1 fans have a fan frame, the fan frame forms a plurality of grooves on two opposite surfaces without any air opening formed, the N+1 fans have a plurality of fixing members disposed in the grooves, and at least one transmission electrode disposed in one of the grooves, the N+1 fans are joined with one another in series through a magnetic attraction effect of the fixing members, the two fan frames of two of the N+1 fans having a joining relationship are flatly contacted with each other and conduction of the two transmission electrodes is completed at a joint between the two N+1 fans, and the N+1 fans transmit electric power through the transmission electrodes. The control source has a connecting seat, a control source electrode disposed on the connecting seat, and a fixing member disposed on the connecting seat, the fixing member of the control source is capable of generating a magnetic attraction effect with one of the fixing members on one of the N+1 fans, when the fixing member of the control source attracts one of the fixing members on one of the N+1 fans, the control source electrode is connected with one of the transmission electrodes.

In one embodiment, each of the transmission electrodes is disposed close to one of the plurality of fixing members.

In one embodiment, the fixing members located on a same surface are disposed at two opposite positions on the surface.

In one embodiment, the N+1 fans are connected in series side by side.

In one embodiment, a side of the connecting seat provided with the fixing member contacts the fan frame flatly.

Through the foregoing implementation of the invention, the invention has the following characteristics compared with the prior art: the invention is implemented joining through a magnetic attraction effect generated by the plurality of fixing members on the N+1 fans, and a connection between the control source and one of the N+1 fans is also implemented by magnetic attraction, which simplify an assembly process.

#### BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a perspective view of a fan set according to a first embodiment of the invention;

FIG. 2 is a perspective view of the fan set according to a second embodiment of the invention;

FIG. 3 is a perspective structural view of a single fan according to the second embodiment of the invention;

FIG. 4 is an exploded perspective view of a fan structure according to the second embodiment of the invention;

FIG. 5 is an exploded perspective view of the fan structure viewed from another angle according to the second embodiment of the invention;

FIG. 6 is a perspective view of the fan set according to a third embodiment of the invention;

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FIG. 7 is a perspective view of the fan set according to a fourth embodiment of the invention;

FIG. 8 is a schematic diagram of the fan set according to a fifth embodiment of the invention;

FIG. 9 is a perspective view of the fan set according to a sixth embodiment of the invention;

FIG. 10 is a perspective view of the fan set according to a seventh embodiment of the invention;

FIG. 11 is a perspective structural view of a single fan according to the seventh embodiment of the invention;

FIG. 12 is an exploded perspective view of a fan structure according to the seventh embodiment of the invention;

FIG. 13 is an exploded perspective view of the fan structure viewed from another angle according to the seventh embodiment of the invention;

FIG. 14 is a perspective view of the fan set according to an eighth embodiment of the invention;

FIG. 15 is a perspective view of the fan set according to a ninth embodiment of the invention;

FIG. 16 is a schematic diagram of the fan set according to a tenth embodiment of the invention; and

FIG. 17 is a schematic diagram of implementation of the invention with a control source.

#### DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

The detailed description and technical contents of the invention are described below with reference to the drawings.

Please refer to FIG. 1. The invention provides a magnetically attracted fan set 100. The fan set 100 comprises N+1 fans, the  $N \geq 1$ . In addition, in order to facilitate the description of the fan set 100 hereinafter, the fan set 100 is temporarily described with the  $N \geq 1$ , that is, the fan set 100 comprises two fans 10, 20.

Each of the fans (10 or 20) has a fan frame (11 or 21) and an airflow generating component (12 or 22) disposed on the fan frame (11 or 21). In one embodiment, the airflow generating component (12 or 22) comprises a driving component (not shown in the figure) disposed inside the fan frame (11 or 21), a fan blade set (121 or 221) disposed inside the fan frame (11 or 21) and connected to the driving component, and a ring (122 or 222) surrounding outside the fan blade set (121 or 221) and defining an airflow channel, the fan blade set (121 or 221) is driven by the driving component to generate an airflow, and the airflow is guided by the ring (122 or 222) during flowing process and flows toward outside of the fan (10 or 20) to which the ring (122 or 222) belongs.

Further, referring to FIG. 1 to FIG. 5, the two fans 10, 20 are disposed adjacent to each other, and the two fans 10, 20 respectively have a joint surface (13 or 23) on adjacent sides. More specifically, each of the fan frames (11 or 21) comprises a top plate (112 or 212) having a first air opening (111 or 211), a bottom plate (114 or 214) parallel to the top plate (112 or 212) and having a second air opening (113 or 213) communicating with the first air opening (111 or 211), and a side wall (115 or 215) connecting the top plate (112 or 212) and the bottom plate (114 or 214). In this embodiment, the joint surface (13 or 23) is a surface of the side wall (115 or 215) of the fan (10 or 20) to which the joint surface (13 or 23) belongs. Practically, the joint surface (13 or 23) can also be the top plate (112 or 212) or the bottom plate (114 or 214) located on the fan (10 or 20) to which the joint surface (13 or 23) belongs, so that the two fans 10, 20 are assembled in a stacking manner. Each of the joint surfaces (13 or 23) is

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provided with at least one fixing member (14 or 24) capable of forming magnetic attraction. For example, the two fixing members 14, 24 of the two fans 10, 20 can each be a magnet (141 or 241), as shown in FIG. 6. In another example, the fixing member (14 or 24) of one of the fans (10 or 20) can be the magnet (141 or 241), and the fixing member (24 or 14) of the other fan (20 or 10) can be a metal member (242 or 142) attracted by the magnet (141 or 241), as shown in FIG. 7. Accordingly, when the two joint surfaces 13, 23 of the two fans 10, 20 are close to each other at a distance sufficient for the two fixing members 14, 24 to form magnetic attraction, the two fans 10, 20 are joined together.

It can be known from the above that the fan set 100 of the invention is provided with the fixing members (14 or 24) on the two fans 10, 20 respectively, so that a user does not need to rely on external components for assembly, and at the same time, if the user loses the external components, a situation of not being able to assemble the conventional fans can be prevented. Furthermore, the two fans 10, 20 of the invention do not need to use specific assembly gestures in assembling, but through magnetic attraction generated by the two fixing members 14, 24 when being close to each other, the two fans 10, 20 can be easily joined together.

Please refer to FIG. 1 to FIG. 4 again. In one embodiment of the invention, in order to assist joining of the two fans 10, 20, one of the fans (10 or 20) has at least one convex portion (15 or 25) or one concave portion (16 or 26), the convex portion (15 or 25) or the concave portion (16 or 26) is located on the joint surface (13 or 23) of the fan (10 or 20) to which the convex portion (15 or 25) or the concave portion (16 or 26) belongs, and the fixing member (14 or 24) is disposed in the convex portion (15 or 25) or the concave portion (16 or 26). The other fan (20 or 10) also has the convex portion (25 or 15) or the concave portion (26 or 16) for connecting with the convex portion (15 or 25) or the concave portion (16 or 26) of the aforementioned fan (10 or 20). In other words, if one of the fans (10 or 20) is provided with the convex portion (15 or 25), the other fan (20 or 10) must be provided with the concave portion (26 or 16). When the two fans 10, 20 are assembled, the convex portion (15 or 25) of one of the fans (10 or 20) is connected with the concave portion (26 or 16) of the other fan (20 or 10).

Further, in the invention, in order to improve a joining stability between the two fans 10, 20, each of the joint surfaces (13 or 23) has the two fixing members (14 or 24), and the two fixing members (14 or 24) are located on opposite sides of the joint surface (13 or 23). In one embodiment, the two fixing members (14 or 24) on the same joint surface (13 or 23) can both be the magnet (141 or 241). In another embodiment, the two fixing members (14 or 24) on one of the joint surfaces (13 or 23) can be the magnet (141 or 241) and the metal member (142 or 242) respectively, and the two fixing members (24 or 14) of the other joint surface (23 or 13) must be the magnet (241 or 141) and the metal member (242 or 142) respectively, and the magnet (241 or 141) of one of the joint surfaces (13 or 23) must be matched with the metal member (242 or 142) on the other joint surface (23 or 13). In this embodiment, each of the fans (10 or 20) can be provided with the convex portion (15 or 25) or the concave portion (16 or 26) to assist the two fans 10, 20 in joining.

On the other hand, please refer to FIG. 1 to FIG. 4 again, each of the fan frames (11 or 21) comprises two half-shells (116 or 216) assembled to each other, and the two half-shells (116 or 216) together form the convex portion (15 or 25) or the concave portion (16 or 26) after assembling. Each of the half-shells (116 or 216) is formed with two accommodating

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grooves (117 or 217) on one side used to form the side wall (115 or 215), and the two accommodating grooves (117 or 217) are located in the convex portion (15 or 25) or in the concave portion (16 or 26), and are provided for disposing the two fixing members (14 or 24) respectively.

In addition, please refer to FIG. 7 and FIG. 8. In one embodiment, each of the fans (10 or 20) can be provided with the joint surface (13 or 23) on two adjacent sides of the side wall (115 or 215) respectively, when the fan set 100 comprises more than three fans (10, 20, 30), the fans (10, 20, 30) can be arranged in a rectangular array or in an L-shaped manner. In another embodiment, each of the fans (10 or 20) can be provided with the joint surface (13 or 23) on two opposite sides of the side wall (115 or 215) respectively, so that the fans (10, 20, 30) of the fan set 100 are arranged in a linear manner. Further, the fans 10, 20, 30 respectively have at least one set of assembling holes (17 or 27 or 37) arranged at two linear ends, the set of assembling holes (17 or 27 or 37) are located at two ends of the fan set 100, the set of assembling holes (17 or 27 or 37) are used to assemble with a set of connectors 400, so that the fan set 100 can be fixed to an electronic device (such as a host computer, not shown in the figures).

On the other hand, please refer to FIG. 9, in one embodiment, the two fans 10, 20 of the fan set 100 of the invention have electrical transmission capabilities. Specifically, each of the joint surfaces (13 or 23) is provided with a transmission electrode (18 or 28), and the transmission electrode (18 or 28) is disposed adjacent to the fixing member (14 or 24). In one embodiment, the transmission electrode (18 or 28) can also be disposed on one of four end corners of the fan (10 or 20) to which the transmission electrode (18 or 28) belongs according to requirements. In another embodiment, the transmission electrode (18 or 28) of one of the fans (10 or 20) can be a transmission pin set, and the transmission electrode (28 or 18) of the other fan (20 or 10) is a transmission socket matched with the transmission pin set of the aforementioned fan (10 or 20). When the two fixing members 14, 24 of the two fans 10, 20 are attracted to each other, the two transmission electrodes 18, 28 of the two fans 10, 20 can be assembled at the same time, so that the two fans 10, 20 directly form an electrical connection relationship.

Accordingly, the invention does not need to be implemented in a way of assembling the electric power and signal transmission pins of the fans after completing mechanical connection of the fans conventionally. In the invention, the two transmission electrodes 18, 28 can be electrically connected at the same time during magnetically attracted assembling of the two fans 10, 20. Furthermore, in the invention, no additional wires are required between the two fans 10, 20 to provide electric power and signal transmission, the two fans 10, 20 can directly transmit electric power and signals through the two transmission electrodes 18, 28 that are in contact with each other, thus unnecessary wires are omitted.

Please refer to FIG. 9 to FIG. 15, in order to expose the transmission electrode (18 or 28) on the fan frame (11 or 21) to which the transmission electrode (18 or 28) belongs, in this embodiment of the invention, each of the half-shells (116 or 216) is formed with a mounting groove (118 or 218) on one side used to form the side wall (115 or 215), the mounting groove (118 or 218) is provided for disposing the transmission electrode (18 or 28), and each of the half-shells (116 or 216) is formed with a notch (119 or 219) corresponding to the mounting groove (118 or 218). After the two half-shells (116 or 216) are assembled, the two mounting grooves 118, 218 jointly limit the transmission electrode (18

or 28), and at the same time the two notches (119 or 219) jointly form an opening (110 or 210) in which the transmission electrode (18 or 28) is exposed.

Further, in one embodiment, each of the fans (10 or 20) is provided with the transmission electrode (18 or 28) on the two joint surfaces (13 or 23) respectively, an electrical conduction relationship is between the two transmission electrodes 18, 28 of the two fans 10, 20, and at least one of the two transmission electrodes (18 or 28) has an electrical conduction relationship with the airflow generating component (12 or 22), so that one of the two transmission electrodes (18 or 28) receives external electric power or signal and transmits it to the airflow generating component (12 or 22).

In another embodiment, please refer to FIGS. 16 and 17, the electronic device can be used as an external electric power or signal source of the fan set 100, and the electronic device is regarded as a control source 500 by the fan set 100. The electronic device has a control source electrode 510, the control source electrode 510 is connected to the transmission electrode (18 or 28) of one of the fans (10 or 20) and transmits electric power to the aforementioned fan (10 or 20). The aforementioned fan (10 or 20) starts after receiving electric power from the electronic device, and transmits the electric power from the electronic device to the other fan (20 or 10), so that the two fans 10, 20 can receive electric power to operate.

In another embodiment, the control source electrode 510 can also transmit signals to the two fans 10, 20, the control source electrode 510 receives a signal transmitted by the control source 500 after being controlled, and transmits the signal to the transmission electrode (18 or 28) of one of the two fans 10, 20 connected to the electronic device. The aforementioned fan (10 or 20) receives the signal from the control source 500 and transmits the signal to the other fan (20 or 10), so that the two fans 10, 20 can be driven by the control source 500 to be capable of producing changes in rotation speed and light.

Further, in this embodiment, when the fan set 100 comprises the fans 10, 20, 30, the control source electrode 510 is connected to the transmission electrode (18 or 28 or 38) of the fans 10, 20, 30 at one end of the fan set 100, after receiving electric power or signal from the control source 500, the aforementioned fan (10 or 20 or 30) transmits it to the fans 10, 20, 30 which are arranged after the aforementioned fan (10 or 20 or 30), so that the fans 10, 20, 30 of the fan set 100 can be driven by the electronic device.

In more detail, in one embodiment, the electronic device has a connecting seat 520 provided with the control source electrode 510, and the connecting seat 520 is provided with a fixing member 522 on a side surface 521 provided with the control source electrode 510. When the control source electrode 510 of the connecting seat 520 is close to the transmission electrode (18 or 28) of one of the fans (10 or 20), the connecting seat 520 generates magnetic attraction with the fixing member (14 or 24) of the aforementioned fan (10 or 20) through the fixing member 522 thereof, so that the connecting seat 520 can be connected to the fan set 100. In addition, in order to improve a stability of assembly between the connecting seat 520 and the fan set 100, the connecting seat 520 of the invention is provided with a convex portion 523 or a concave portion (not shown in the figures) at a position of the fixing member 522 according to requirements, so that the convex portion 523 can be connected with the concave portion (16 or 26) of the aforementioned fan (10 or 20).

Note that the specification relating to the above embodiments should be construed as exemplary rather than as limitative of the present invention, with many variations and modifications being readily attainable by a person of average skill in the art without departing from the spirit or scope thereof as defined by the appended claims and their legal equivalents.

What is claimed is:

1. A magnetically attracted fan set, comprising,

N+1 fans, wherein  $N \geq 1$ , each of the N+1 fans comprises a plurality of fixing members for joining, and a plurality of transmission electrodes respectively located on a surface as same as at least one of the plurality of fixing members, the N+1 fans are joined with one another through magnetic attraction by the plurality of fixing members, and one of the plurality of transmission electrodes of each of the N+1 fans is connected to one of the plurality of transmission electrodes of the other of the N+1 fans to transmit electric power; and

a control source comprising a connecting seat, a control source electrode disposed on the connecting seat, and a fixing member disposed on a side of the connecting seat provided with the control source electrode, and a magnetic attraction effect being generated by the fixing member, when the fixing member and the plurality of fixing members of one of the N+1 fans generating a magnetic attraction effect, the control source electrode being connected to one of the plurality of transmission electrodes,

wherein each of the plurality of transmission electrodes is capable of being connected to one of the plurality of transmission electrodes of the other of the N+1 fans and being connected to the control source electrode.

2. The magnetically attracted fan set as claimed in claim 1, wherein each of the plurality of transmission electrodes is disposed adjacent to one of the plurality of fixing members on the same surface.

3. The magnetically attracted fan set as claimed in claim 1, wherein each of the N+1 fans comprises two transmission electrodes, and the two transmission electrodes are respectively located on two sides of each of the N+1 fans opposite to each other.

4. The magnetically attracted fan set as claimed in claim 3, wherein the plurality of fixing members located on one surface are located at two positions on the surface opposite to each other.

5. The magnetically attracted fan set as claimed in claim 4, wherein the N+1 fans are connected in series side by side.

6. The magnetically attracted fan set as claimed in claim 5, wherein the N+1 fans comprise a fan frame provided for disposing of the plurality of fixing members and the plurality of transmission electrodes, the fan frame comprises a top plate comprising a first air opening, a bottom plate parallel to the top plate and comprising a second air opening communicating with the first air opening, and a side wall connecting the top plate and the bottom plate, the side wall is formed with a plurality of grooves provided for disposing of the plurality of fixing members and the plurality of transmission electrodes.

7. The magnetically attracted fan set as claimed in claim 6, wherein a surface of the connecting seat provided with the fixing member contacts the fan frame flatly.

8. A magnetically attracted fan set, comprising:

N+1 fans, wherein  $N \geq 1$ , and the N+1 fans comprise a fan frame forming a plurality of grooves on two surfaces opposite to each other, and each of the two surfaces is without an air opening formed, the N+1 fans comprise

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a plurality of fixing members disposed in the plurality of grooves, and at least one of plurality of transmission electrodes disposed in one of the plurality of grooves, the N+1 fans are joined with one another through a magnetic attraction effect generated by the plurality of fixing members, the N+1 fans transmit electric power through the plurality of transmission electrodes; and

a control source comprising a connecting seat, a control source electrode disposed on the connecting seat, and a fixing member disposed on the connecting seat, the fixing member of the control source generating a magnetic attraction effect with one of the plurality of fixing members on one of the N+1 fans, when the fixing member of the control source attracting one of the plurality of fixing members on one of the N+1 fans, the control source electrode being connected with one of the plurality of transmission electrodes,

wherein each of the plurality of transmission electrodes is capable of being connected to one of the plurality of transmission electrodes of the other of the N+1 fans and being connected to the control source electrode.

9. The magnetically attracted fan set as claimed in claim 8, wherein each of the plurality of transmission electrodes is disposed adjacent to one of the plurality of fixing members on the same surface.

10. The magnetically attracted fan set as claimed in claim 9, wherein the plurality of fixing members on one surface are located at two positions of the surface opposite to each other.

11. The magnetically attracted fan set as claimed in claim 10, wherein the N+1 fans are connected in series side by side.

12. The magnetically attracted fan set as claimed in claim 9, wherein a surface of the connecting seat provided with the fixing member contacts the fan frame flatly.

13. A magnetically attracted fan set, comprising: N+1 fans, wherein  $N \geq 1$ , and the N+1 fans comprise a fan frame forming a plurality of grooves on two surfaces opposite to each other, and each of the two surfaces is without an air opening formed, the N+1 fans comprise a plurality of fixing members disposed in the plurality

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of grooves, and at least one of plurality of transmission electrodes disposed in one of the plurality of grooves, the N+1 fans are joined with one another in series through a magnetic attraction effect generated by the plurality of fixing members, two fan frames of two of the N+1 fans with a joining relationship are flatly contacted with each other and completion of conduction of two transmission electrodes at a joint therebetween, and the N+1 fans transmit electric power through the plurality of transmission electrodes; and

a control source comprising a connecting seat, a control source electrode disposed on the connecting seat, and a fixing member disposed on the connecting seat, the fixing member of the control source generating a magnetic attraction effect with one of the fixing members on one of the N+1 fans, when the fixing member of the control source attracting one of the plurality of fixing members on one of the N+1 fans, the control source electrode being connected with one of the plurality of transmission electrodes,

wherein each of the plurality of transmission electrodes is capable of being connected to one of the plurality of transmission electrodes of the other of the N+1 fans and being connected to the control source electrode.

14. The magnetically attracted fan set as claimed in claim 13, wherein each of the plurality of transmission electrodes is disposed adjacent to one of the plurality of fixing members on the same surface.

15. The magnetically attracted fan set as claimed in claim 14, wherein the plurality of fixing members on one surface are located at two positions of the surface opposite to each other.

16. The magnetically attracted fan set as claimed in claim 15, wherein the N+1 fans are connected in series side by side.

17. The magnetically attracted fan set as claimed in claim 16, wherein a surface of the connecting seat provided with the fixing member contacts the fan frame flatly.

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